Table 1: Overall contact fidelity and dynamic quality on CORE4D-G1 (52 sequences / 11 objects). ↑: higher is better, ↓: lower is better. **Bold** = best, underline = second best. Ours = SPIDER-CEM. Contact metrics are penetration-aware (a loose "any-contact" count that rewards interpenetration is intentionally omitted). Dynamic-quality metrics for Ours, SBTO and OmniRetarget are computed on a MuJoCo physics rollout (native for the two optimizers; position-servo replay for the kinematic OmniRetarget, E197 protocol) via the same code, hence directly comparable; GMR's are from its own kinematic pipeline at a different scale and are indicative only (read jointly with contact).

Metric	Ours	SBTO	OmniRt	GMR
Phys. contact@3mm ↑	**0.588**	0.190	0.101	0.004
Phys. contact@10mm ↑	**0.804**	0.650	0.510	0.014
Phys. pen@3mm ↓	**0.176**	0.490	0.536	0.349
Phys. pen@10mm ↓	**0.017**	0.136	0.247	0.339
Max phys pen (mm) ↓	**12.3**	25.1	21.1	102.1
Geom. pen@2mm ↓	**0.052**	0.425	0.381	0.456
Ankle jerk p95 (m/s^3) ↓	653.8	1041.7	1560.8	**170.4**
Obj speed max (m/s) ↓	**1.29**	1.77	2.83	1.86
Foot slip max (m) ↓	**0.87**	1.20	1.33	0.96